

tata retail project

This analysis focuses on extracting meaningful business insights from an online retail dataset to support data-driven decision-making. The project addressed four key business questions and aimed to provide clear, actionable insights for leadership.

The first step involved data preparation and quality assurance. The dataset included transactional records such as invoices, product details, quantities, prices, customer IDs, and geographic information. Three major data quality issues were identified and resolved.

Records with negative quantities, representing returns or cancellations, were removed to ensure accurate revenue analysis. Rows with zero or negative unit prices were excluded, as they did not represent valid sales transactions. Additionally, records with missing customer IDs were removed to maintain accuracy in customer-level analysis. This cleaning process ensured that all insights were based on reliable and validated data.

Data Cleaning Process for Business Insights



Remove Negative Quantities

Eliminate records representing returns or cancellations



Exclude Zero/Negative Prices

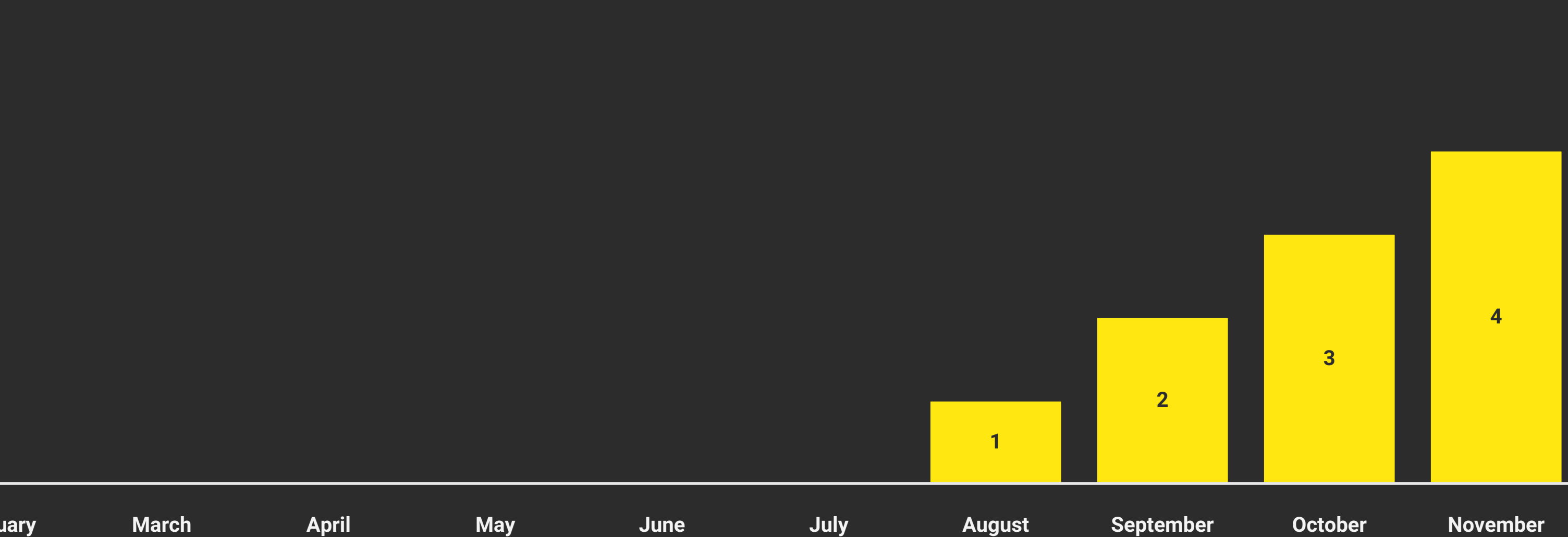
Remove records that are not valid sales transactions



Remove Missing Customer IDs

Ensure accuracy in customer-level analysis

The analysis of monthly revenue trends for 2011 revealed a clear seasonal pattern. Revenue remained stable from January to July, followed by a steady increase starting in August. The growth accelerated significantly, reaching a peak in November, with a slight decline in December. This pattern indicates strong customer purchasing behavior during the holiday season. From a business perspective, this highlights a strategic opportunity to focus on the August to November period by optimizing inventory, increasing marketing efforts, and launching targeted promotional campaigns to maximize revenue.



Monthly Revenue Trends in 2011

The evaluation of international markets, excluding the United Kingdom, provided insights into global performance. The Netherlands emerged as the top-performing country, followed by Ireland, Germany, and France. Other notable contributors included Australia, Spain, Switzerland, Belgium, Sweden, and Japan. The results clearly indicate that Europe is the strongest international region, with multiple high-performing markets. A focused expansion strategy in Europe is recommended, as it offers established demand and lower risk. Australia also presents a promising growth opportunity due to its strong performance despite geographic distance.

Top International Market Performers



The analysis of the top 10 customers by revenue showed that while the highest-spending customer contributes significantly, the overall revenue distribution is relatively balanced. This indicates that the business is not overly dependent on a single customer, reducing potential risk. High-value customers should be prioritized through personalized engagement, loyalty programs, and exclusive offers, as retaining these customers can have a significant impact on overall revenue.

Top Customer Revenue Distribution

Customer Retention

Retaining high-value customers boosts revenue.

Highest-Spending Customer

Contributes significantly to overall revenue.

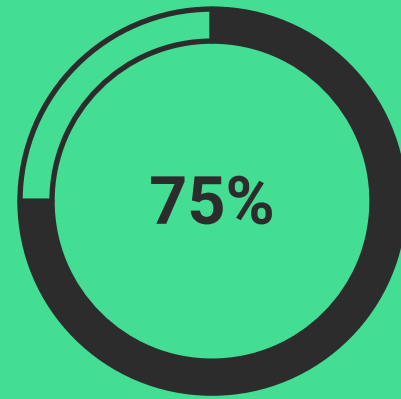
Balanced Revenue Distribution

Revenue is relatively balanced among top customers.



Finally, the geographic demand analysis highlighted strong demand concentration in Western Europe, particularly in the Netherlands, Germany, France, and Ireland. Moderate demand was observed in Australia and parts of Asia. These insights reinforce the strategic importance of focusing on European markets for expansion while considering Australia as a secondary growth opportunity.

Geographic Demand Analysis



Western Europe Demand

High demand in Netherlands, Germany, France, Ireland.



Australia and Asia Demand

Moderate demand in Australia and parts of Asia.

Western Europe shows the strongest demand, making it the primary focus for expansion.

Overall, the analysis identifies key opportunities in seasonal demand, international expansion, customer retention, and geographic targeting. Aligning business strategies with these insights will support sustainable growth and improved performance.